CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 9: Matrices

Previous year board questions, topic-wise: 14 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Multiplication of matrices

1. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, find A^2. (a) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ICSE 2025, 1 mark

2. Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} -5 & 1 \\ 7 & -4 \end{bmatrix}$, find (a) $A + C$ (b) $B(A + C)$ (c) $5B$ (d) $B(A + C) - 5B$. ICSE 2025, 4 marks

3. If $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$ and $A^2 = \begin{bmatrix} 4 & x \\ 0 & 4 \end{bmatrix}$, find x. (a) 2 (b) 4 (c) 8 (d) 10 ICSE 2024, 1 mark

4. Given $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 1 \\ 1 & 5 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, find $A(B + C) - 14I$. ICSE 2023, 3 marks

5. If $A = \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 2 \\ 1 & 0 \end{bmatrix}$, find $A^2 - 2AB + B^2$. ICSE 2020, 3 marks

6. If $A = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ -1 & 7 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 0 \\ -1 & 4 \end{bmatrix}$, find $AC + B^2 - 10C$. ICSE 2018, 3 marks

7. $A = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -2 & 1 \\ -3 & 2 \end{bmatrix}$ and $A^2 - 5B^2 = 5C$. Find the matrix C. ICSE 2017, 4 marks

2 Solving matrix equations

8. $A = \begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 0 \\ y & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 0 \\ x & 1 \end{bmatrix}$. Find x and y if $AB = C$. ICSE 2024, 4 marks

9. If $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}$, find x and y. (a) $1, -2$ (b) $-2, 1$ (c) $1, 2$ (d) $-2, -1$ ICSE 2023, 1 mark

10. $A = \begin{bmatrix} x & 3 \\ y & 3 \end{bmatrix}$ and $A^2 = 3I$, where I is the identity matrix of order 2. Find x and y. ICSE 2020, 3 marks

11. Given $\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} M = 6I$, where I is the unit matrix of order 2×2, (a) state the order of M (b) find M. ICSE 2019, 4 marks

12. Given $B = \begin{bmatrix} 1 & 1 \\ 8 & 3 \end{bmatrix}$, find $X = B^2 - 4B$. Then solve $X \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5 \\ 50 \end{bmatrix}$ for a and b. ICSE 2017, 4 marks

3 Addition, subtraction and multiplication by a number

13. Simplify $\sin A \begin{bmatrix} \sin A & -\cos A \\ \cos A & \sin A \end{bmatrix} + \cos A \begin{bmatrix} \cos A & \sin A \\ -\sin A & \cos A \end{bmatrix}$. ICSE 2019, 3 marks

14. Find x and y if $2 \begin{bmatrix} x & 7 \\ 9 & y-5 \end{bmatrix} + \begin{bmatrix} 6 & -7 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 10 & 7 \\ 22 & 15 \end{bmatrix}$. ICSE 2018, 3 marks

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